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Cosmic Diaspora Hypothesis: A Structural Analysis Framework on Cosmic Life Diffusion, Civilization Evolution, and Extraterrestrial Phenomena

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This paper proposes a structural hypothesis framework addressing three interconnected themes: under the premise that the abiogenesis probability is not extremely low, the diffusion of life across the universe as a statistically high-probability outcome rather than a rare event; the deconstruction of contemporary "alien" phenomena as a projected ensemble of multiple possible sources rather than a single-species narrative; and the derivation of plausible physical and engineering characteristics for advanced civilization base sites, stealth strategies, and energy technologies under the conditional assumption of existence. The framework is built upon extensions of known physics—including statistical cosmology, materials science, energy hierarchies, and electromagnetic signature control—without presupposing the existence of extraterrestrial civilizations or violating fundamental physical laws. A multi-source classification model decomposes extraterrestrial-related phenomena into independent categories including military misjudgment, sensor anomalies, extra-solar observers, intra-solar-system outposts, ancient Martian and Venusian civilization remnants, ancient Earth civilization continuations, and non-biological intelligence probes. The hypothesis further develops a cross-galactic deployment architecture in which civilization migrates sequentially among solar system planets as habitability windows shift, with Mars, Venus, and Earth forming a temporal relay chain. Specific falsifiable observation pathways—including subsurface imaging, deep-sea scanning, gravitational anomaly monitoring, electromagnetic spectrum analysis, and planetary exploration—are proposed to enable empirical evaluation of each structural model.

I. METHODOLOGY & SCOPE

A. Problem Definition

This hypothesis attempts to answer the following core questions:

- Is the emergence of life in the universe a rare exception or a statistical inevitability?
- If advanced civilizations exist, can their technological maturation pathway be derived from known physics?
- Can contemporary "alien" phenomena be deconstructed through a multi-source classification model?
- If extraterrestrial bases exist, do their engineering characteristics produce observable physical signals?

B. Physics & Engineering Feasibility Derivation Principles

This hypothesis adheres to the following constraints:

- **No violation of known physical laws**: All derivations are built upon reasonable extensions of the existing physics framework.
- **Engineering feasibility orientation**: Technological assumptions must have precursor foundations in current scientific research (e.g., fusion energy research, quantum computing development, metamaterial research). Program-level references (ITER [1], IBM quantum hardware [2], SpaceX [3], NASA Artemis [4], Long Now Foundation [5]) are cited as contextual indicators of engineering direction, not as evidential support for specific claims.
- **Incremental extrapolation principle**: Deriving possible evolutionary paths thousands to millions of years beyond current human technological levels.

C. Statistical Cosmology Framework

This hypothesis is based on the following statistical premises:

- The Milky Way contains approximately 100–400 billion stars.
- Exoplanet observation missions (e.g., Kepler, TESS) demonstrate that planet formation is a normative process.

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- On cosmic timescales (10^9–10^{10} year scale), low-probability events can accumulate into high aggregate probabilities.

Observation Selection Effects Statement: This framework's statistical derivations require an additional epistemological caveat: the Anthropic Principle imposes non-trivial constraints on probabilistic reasoning. As observers, our very existence pre-selects an environment in which life has already emerged—this means we cannot directly infer "life is common in the universe" from the fact that "life exists on Earth" (Bostrom, 2002 [6]).

Specifically: even if life is extraordinarily rare in the universe ($f_l \to 0$), as long as at least one civilization emerges and produces observers, those observers will necessarily observe that their planet harbors life. This self-selection bias means that an $N = 1$ observation cannot constrain the lower bound of f_l.

Critically, Sandberg, Drexler & Ord (2018) [7] demonstrated that when Drake Equation parameters are treated as realistic probability distributions rather than point estimates, the Fermi Paradox can be dissolved without invoking additional complex mechanisms—the probability of humanity being alone in the observable universe is non-negligible. This result directly constrains the necessity of frameworks that posit hidden civilizations: simpler explanations (we are rare or alone) must be given appropriate prior weight. This framework's conditional structure ("if civilizations exist, then...") is designed to remain valid regardless of how this prior resolves.

This framework acknowledges this limitation and positions it as supplementary background for Sec. II A 6 (Abiogenesis Constraint). The framework's core derivations (Parts II through VIII) employ conditional structure ("if life/civilization already exists, then..."), and therefore are not affected by anthropic bias.

D. Phenomenon–Structure Mapping

For each extraterrestrial-related phenomenon, this hypothesis does not directly judge "true or false" but instead:

1. Enumerates all possible physical source models.
2. Evaluates the technological prerequisites required by each model.
3. Determines whether those prerequisites fall within the extension range of known physics.
4. Constructs a "phenomenon → structure" correspondence table, replacing narrative-based judgment.

E. Falsifiability & Boundary Conditions

This hypothesis is designed as a falsifiable framework:

- **If high-resolution subsurface imaging fails to detect anomalous structures**, then underground base models can be weakened.
- **If comprehensive deep-sea scanning reveals no artificial features**, then oceanic base models can be eliminated.
- **If gravitational anomaly monitoring cannot identify non-natural distribution patterns**, then base energy models require revision.
- **If future materials science proves room-temperature superconductivity infeasible**, then related technological derivations need adjustment.

This hypothesis does not claim "aliens exist"—it provides an analytical tool for "if they exist, what derivable structures should they exhibit."

Evidence Standards: This framework adopts the general principle articulated in the NASA/NfoLD community life-detection framework [8]: claims of detection require progression through multiple evidence levels—from scenario generation and observation-path suggestion, through signal identification, non-biological source exclusion, to convergence of independent evidence lines. The sub-models in Parts III–VIII currently occupy the earliest stages (scenario generation and observation-path suggestion); none have yet reached the signal-identification level. This self-assessment should guide interpretation of all claims herein.

Epistemological Layering of the Framework: This hypothesis contains two levels with distinct epistemological statuses:

1. **Falsifiable sub-models** (underground bases, undersea bases, Venus atmospheric habitation, etc.)—each sub-model defines specific weakening or exclusion conditions (see Sec. VIII A). The scientific value of these sub-models lies in their capacity to be negated by empirical evidence.
2. **Non-falsifiable meta-framework** (distributed non-material civilization forms, see Sec. III B 6)—this form assumes civilizations leave no physical traces whatsoever, and therefore cannot in principle be excluded by current observational technology. This hypothesis explicitly acknowledges: this meta-framework exceeds the boundaries of falsifiable science; its status is that of a purely speculative thought experiment and should not be evaluated on the same terms as falsifiable sub-models.

The observation pathways in Part VIII apply only to falsifiable sub-models. If all falsifiable sub-models

are excluded by empirical evidence, the distributed non-material civilization hypothesis may still exist as a logical possibility, but its scientific status will be equivalent to a non-falsifiable metaphysical proposition.

II. PART I — FOUNDATIONS OF COSMIC LIFE DIFFUSION

A. Statistical Foundations of Life in the Universe

1. *Stellar & Planetary Formation as Normality*

Stellar formation rates and planetary disk generation are natural processes of celestial evolution. Based on current observational data:

- The Milky Way's star formation rate is approximately 2 solar masses per year (corresponding to roughly 6–7 individual stars, most being low-mass red dwarfs) [9].
- Planetary system formation is a natural byproduct of the star formation process.
- Over the universe's 13.8-billion-year history, the accumulated number of stars and planets is extraordinarily vast.

Planet formation is not a special event but a statistical norm of astrophysics.

2. *Earth-like Planets & Habitable Zone Distribution*

Based on exoplanet observation missions (e.g., Kepler Space Telescope [10], TESS [11]):

- The estimated number of Earth-like planets in the Milky Way reaches billions to approximately 40 billion (depending on definitional scope and stellar types included) [12].
- Habitable Zone formation is a natural byproduct of orbital dynamics.
- The probability of liquid water existence is statistically significant.

On cosmic timescales (10^9 year scale):

- Earth-like planet formation is normative.
- Water presence probability is significant.
- Conditions for the origin of life fall within the statistically reachable range.

3. *Ubiquity of Organic Molecules in the Universe*

Over 300 molecular species have been detected in the interstellar medium as of 2025 [13] (the vast majority being organic), including:

- Amino acids and their precursors (e.g., glycine detected toward a solar-type protostar; see [14], a preprint whose DOI resolves to first author Quan rather than Rivilla—citation metadata may change upon formal publication)
- Polycyclic aromatic hydrocarbons (PAHs)
- Alcohols, aldehydes, and nitrogen-containing organics

Meteorite samples (e.g., Murchison meteorite) have yielded over 90 types of amino acids (approximately 96 confirmed as of 2017) [15]. This demonstrates that organic chemistry is a universal process throughout the cosmos, not a phenomenon unique to Earth.

4. *Extremophile Life Forms*

Extremophiles on Earth have demonstrated that life can survive under the following conditions:

The existence of these organisms dramatically expands the definitional boundaries of "habitability."

5. *Interplanetary Lithopanspermia & Temporal Accumulation Effects*

The physical mechanisms of **Panspermia** [18]:

1. **Impact ejection**: Asteroid or comet impacts can launch microbe-bearing rocks into space.
2. **Space survival**: Certain extremophiles (e.g., *Deinococcus radiodurans*) possess long-term survival capabilities in vacuum, intense radiation, and extreme cold environments.
3. **Re-entry survival**: Rock interiors can provide sufficient thermal shielding for microorganisms to survive atmospheric re-entry.

Panspermia does not violate known physical laws. On cosmic lifespan timescales (10^{10} year scale), while individual transfer probabilities are low:

- Cumulative aggregate probability rises significantly over time.
- Distances between stellar systems within the Milky Way are not insurmountable.
- Life diffusion may be a slow but continuous cosmic-scale process.

TABLE I. Extremophile organisms and their tolerance ranges.

Environment	Extreme Condition	Known Organism Example
High Temperature	>120 °C	Hyperthermophilic archaea (e.g., *Methanopyrus kandleri*) [16]
Low Temperature	<−20 °C	Antarctic ice sheet microorganisms
High Pressure	>1000 atm	Deep-sea trench organisms (e.g., Mariana Trench microbes)
High Radiation	>5000 Gy	*Deinococcus radiodurans* [17]
High Salinity	Saturated brine	*Halobacterium* spp.
Anoxic	Completely anaerobic	Methanogens

6. *The Unresolved Constraint of Abiogenesis Probability*

The preceding sections have established the following evidence chain: stellar and planetary formation as normality (Sec. II A 1) → widespread habitable zone distribution (Sec. II A 2) → universal presence of organic molecules (Sec. II A 3) → life can survive in extreme environments (Sec. II A 4) → interplanetary rock transfer is physically feasible (Sec. II A 5).

However, a critical break point exists in this evidence chain: **the transition probability from organic molecules to the first self-replicating system (abiogenesis) is currently completely unknown.** This constitutes the most fundamental source of uncertainty in this framework.

Specifically:

- The existence of organic precursors (amino acids, nucleotides, etc.) is a necessary but not sufficient condition for life.
- To date, no laboratory experiment has successfully produced a complete self-replicating system spontaneously from an inorganic environment.
- Earth is the only known data point for the origin of life ($N = 1$), which means we cannot make meaningful statistical inferences about the abiogenesis probability (f_l in the Drake Equation)—its value could range from $\sim$1 (near-certain) to 10^{-40} (extraordinarily rare). A Bayesian analysis by Lingam et al. (2024) [19] provides a quantitative framework for this uncertainty, demonstrating that the posterior distribution of abiogenesis probability per conducive site remains broad and does not yet favor either extreme.
- Panspermia (Sec. II A 5) does not resolve this problem; it merely relocates the site of abiogenesis to another celestial body.

Impact of this constraint on the framework: If the true value of f_l is extremely low (e.g., $< 10^{-20}$), then even with hundreds of billions of habitable planets in the universe, life may still be an extraordinarily rare event. In this scenario, Part I's conclusion of "high-probability statistical outcome" would not hold, but the conditional derivations in Parts II through VIII ("if life/civilization already exists, then...") remain valid, as they do not depend on the specific value of f_l.

This framework explicitly acknowledges abiogenesis as an unresolved constraint and marks it as the single link with the highest uncertainty in the overall derivation chain.

III. PART II — CIVILIZATION EVOLUTION & TECHNOLOGY THRESHOLDS

A. Energy & Material Thresholds of Civilization Evolution

1. *Energy Hierarchy Model*

A civilization's technological level is directly correlated with its energy mastery capabilities. Extending the Kardashev Scale [20]:

Key observation: Every energy-level transition is accompanied by a fundamental breakthrough in materials science. Increasing energy density is not merely a physics problem—it is a materials engineering problem.

2. *Material Breakthrough Conditions*

The bottleneck for civilization's technological leaps often lies not in theoretical physics but in materials science:

- **Fusion bottleneck**: Not the fusion reaction itself, but the containment vessel materials that must withstand ultra-high-temperature plasma.
- **Quantum computing bottleneck**: Not quantum gate design, but the material environment needed to maintain quantum coherence.
- **Space travel bottleneck**: Not propulsion theory, but structural materials resistant to extreme environments.

Each breakthrough in materials science unlocks an entire domain of engineering possibilities.

TABLE II. Energy hierarchy model extending the Kardashev Scale.

Level	Energy Scale	Human Reference Point
Type 0.7	$\sim 10^{13}$ W	Current human civilization ($\sim$19 TW)
Type I	$\sim 10^{16}$–10^{17} W	Harnessing entire planetary energy output
Type II	$\sim 10^{26}$ W	Harnessing entire stellar energy output
Type III	$\sim 10^{36}$ W	Harnessing entire galactic energy output

3. *Superconductors & Extreme Materials*

Core physical properties of superconducting materials [21]:

- **Zero resistance**: Lossless current transmission.
- **Meissner Effect** [22]: Complete expulsion of external magnetic fields.
- **Extremely high current density**: Capable of transmitting enormous currents through minimal cross-sectional areas.

The current limitation of human superconductor technology lies in critical temperature. If a civilization has mastered room-temperature, ambient-pressure superconducting materials, the following technologies become immediately viable:

1. **Lossless long-distance energy transmission**—Energy system efficiency approaches theoretical limits.
2. **High-efficiency magnetic plasma confinement**—Fusion reactor volumes can be drastically reduced.
3. **Stable magnetic levitation structures**—Fundamental transformation of transportation and construction engineering.
4. **Ultra-low thermal dissipation energy systems**—Facilities exhibit virtually no external thermal signatures.

4. *Quantum Computing & Higher-Dimensional Simulation*

Companies like IBM have already constructed superconducting quantum processors [2], but current qubit coherence times are extremely short and error rates remain high.

If a civilization masters:

- **Topological Qubits**—Leveraging topological protection to naturally resist environmental interference.
- **Room-temperature quantum coherence materials**—Eliminating cryogenic cooling requirements.
- **Long-duration entanglement maintenance structures**—Making large-scale quantum computation stably feasible.

Then they could achieve:

- Simulating planetary climate and ecological evolution.
- Optimizing base locations and stealth parameters.
- Precisely predicting energy flows and gravitational anomalies.
- Conducting higher-dimensional complex system simulations.

5. *Biological–Mechanical Integration Possibilities*

If a civilization possesses the following material foundations:

- **Bioconductive material integration**—Seamless interfaces between organic bodies and electronic systems.
- **Neural-electromagnetic interface materials**—Direct read/write access to neural signals.
- **Controlled nanoscale energy input structures**—Precise energy delivery at the cellular level.
- **Long-range energy receiver antenna structures**—Miniature energy reception arrays embedded in biological bodies, capable of wirelessly receiving energy from remote energy nodes.

Then biological organisms could become:

- **Rechargeable systems**—Not entirely dependent on metabolic food intake; can be continuously replenished via long-range wireless energy transmission.
- **Semi-self-sustaining energy nodes**—Harvesting energy from environmental fields or directed energy beams.
- **Distributed computing units**—Collaborative computation between biological brains and quantum processors.

- **Remotely operable carriers**—Receiving remote commands via brain-computer interfaces; individuals serve as remote extension terminals of the civilization network.

Key derivations for brain-computer interfaces and remote operation:

If brain-computer interface technology matures to the point of bidirectional transmission of complete neural signals, then:

1. **Remote operation of biological bodies becomes possible**—Central nodes or other individuals can send commands directly to the target body's motor cortex and sensory cortex via quantum-encrypted channels, enabling precise remote control.
2. **Shared perception networks**—Multiple individuals can simultaneously share visual, auditory, and tactile information, forming distributed perceptual coverage.
3. **Consciousness-level information transmission**—Transcending linguistic symbols to directly transmit concepts, images, and emotions; communication bandwidth can reach several orders of magnitude above electromagnetic wave communication.
4. **Decentralized decision-making**—Each individual serves as both sensor and actuator; the civilization as a whole forms a neural-network-like distributed intelligence.

This derivation does not violate bioelectrochemical principles—it only requires materials engineering breakthroughs. Key supporting technologies include:

- Highly stable superconducting biological interfaces
- Biocompatible nanomaterials
- Precise quantum-level signal control
- Quantum-secured communication channels (utilizing quantum key distribution to ensure communication security, paired with highly directional photon carriers, making signals practically undetectable by third parties)
- Directed wireless energy transmission systems (microwave or laser energy beams)

B. Advanced Civilization Technology Derivation Model

1. *Compact Nuclear Fusion & Material Limits*

ITER (International Thermonuclear Experimental Reactor) [1] currently faces a core bottleneck that **is not theoretical feasibility, but material endurance limits**:

- **High-energy neutron bombardment**: D-T fusion produces 14.1 MeV neutrons that cause sustained radiation damage to vessel walls.
- **Ultra-high-temperature plasma containment**: Plasma temperatures reach 150 million°C, far exceeding the direct contact tolerance of any known material.
- **Strong magnetic field stability requirements**: Tokamaks require stable magnetic fields on the order of several Tesla to confine plasma.

If an advanced civilization has matured the following material technologies:

- **High-Entropy Alloys (HEAs)** [23]—Multi-principal-element alloy systems with exceptional high-temperature strength, radiation damage resistance, and corrosion resistance.
- **Radiation self-healing materials**—Crystal structures with atomic-level self-repair mechanisms that autonomously restore lattice defects under neutron bombardment.
- **Neutron absorption-optimized lattice structures**—Precisely engineered lattice geometries that maximize neutron kinetic energy absorption and conversion efficiency.

Then compact fusion reactors become engineering-feasible. This means base energy systems can:

- **Operate in long-term isolation**—If a complete tritium breeding cycle is established, long-term closed operation becomes feasible—deuterium can be extracted almost limitlessly from seawater, and tritium is self-produced via lithium breeding blankets.
- **Leave virtually no external energy signature**—No smokestacks, no cooling towers, no large-scale grid connections.
- **Achieve energy density far exceeding fossil and fission technologies**—A single reactor can power an entire base for decades.

Fuel Cycle Constraint:

Note that the fuel self-sufficiency claim for D-T fusion requires qualification:

- **Deuterium (D)**: Can be extracted from seawater in virtually unlimited quantities (seawater deuterium content $\sim$33 mg/L).

- **Tritium (T)**: Has a half-life of only 12.3 years and is virtually non-existent in nature. Reactors must self-produce tritium through a Lithium Breeding Blanket (^{6}Li+$n \rightarrow T+^4$He). The Tritium Breeding Ratio (TBR) must be > 1.0 to maintain fuel self-sufficiency, imposing rigid requirements on blanket volume and neutronic design—the reactor has a minimum size floor determined by the breeding blanket volume, which constitutes a physical constraint on miniaturization.

Alternative fuel pathways (D-D fusion or p-^{11}B fusion) can avoid tritium dependence, but require higher plasma temperatures (D-D: ∼300–500 million K; p-^{11}B: ∼1–3 billion K) and have lower reaction cross-sections, further escalating the difficulty of materials and confinement engineering. An advanced civilization's choice of fusion technology pathway will directly determine the minimum viable reactor size.

2. *Room-Temperature Superconducting Materials & Magnetic Field Manipulation*

The role of superconducting materials in base engineering extends beyond energy transmission—they form the architectural foundation of the entire system:

Energy system applications:

In terms of base stealth:

- **No significant heat emissions**—Superconducting systems produce no Joule heating.
- **No detectable electromagnetic leakage**—Superconducting shielding can completely confine magnetic fields within the system.
- **Extremely low acoustic signature**—No rotating generators, no cooling fans, no mechanical vibration.

Such technologies enable underground or undersea facilities to achieve long-term low-signature operation.

3. *Topological & Metamaterials*

a. Topological Materials. Key physical properties of topological materials [24]:

- **Surface-conducting, internally insulating**—Electron conduction occurs only on the material surface and is topologically protected.
- **Insensitive to defects and impurities**—Topological states are unaffected by local perturbations, possessing natural anti-interference capabilities.
- **Spin-momentum locking**—Electron spin direction is fixedly coupled to its direction of motion.

Application derivations:

- **Highly stable quantum computing cores**—Topological protection makes qubits naturally immune to most decoherence mechanisms.
- **Anti-interference energy channels**—Energy transmission paths unaffected by material defects.
- **High-sensitivity sensors**—Extreme sensitivity of surface states to external fields.

b. Metamaterials. Metamaterials are artificially designed structured materials whose electromagnetic properties are determined by geometric structure rather than chemical composition [25]:

- **Designable negative refractive index**—Light can be guided along non-natural propagation paths.
- **Controllable electromagnetic wave paths**—Enabling arbitrary bending and redirection of electromagnetic waves.
- **Electromagnetic cloaking effects**—Making objects "transparent" to specific frequency bands of electromagnetic waves.

Base stealth derivations:

- **Radar invisibility**—Guiding incident radar waves around the structure body, rendering it invisible on radar.
- **Electromagnetic wave scatter redirection**—Redirecting detection signals in other directions, creating false echoes.
- **Thermal infrared signature suppression**—Controlling the directionality and spectral distribution of thermal radiation, reducing infrared visibility.

If metamaterial engineering is matured, bases can physically and substantially reduce their detectability.

c. High-Strength Carbon-Based & Nanostructured Materials. Representative materials:

- **Multi-layer graphene stacks**—Ordered stacking structures of single-atom-thick carbon layers.
- **Carbon nanotube composites**—Hollow carbon tubes with extreme aspect ratios.

Core physical properties:

In deep-sea or polar pressure environments:

- **Significantly enhanced structural stability**—Carbon-based materials have strength-to-weight ratios far exceeding traditional metals.

TABLE III. Superconductor applications in energy systems.

Application	Superconductor Advantage	Engineering Significance
Energy transmission	Zero resistance	Theoretically unlimited transmission distance, no thermal losses
Fusion magnetic field	Extremely high current density	Reactor volume reduction by 1–2 orders of magnitude
Energy storage	SMES	Instantaneous high-power release, no chemical degradation
Magnetic levitation	Meissner Effect	Contactless transport systems, zero friction loss

TABLE IV. Mechanical properties of carbon-based nanomaterials.

Property	Graphene	Carbon Nanotubes
Tensile Strength	∼130 GPa [26]	∼60 GPa [27]
Young's Modulus	∼1 TPa [26]	∼1 TPa [27]
Thermal Conductivity	∼5000 W/mK	∼3500 W/mK
Electron Mobility	Extremely high	Chirality-dependent

- **Improved energy and thermal management efficiency**—Extremely high thermal conductivity allows waste heat to be precisely directed and processed.

- **Lightweight advantage**—At equivalent strength, structural mass can be reduced by 1–2 orders of magnitude.

d. Material System Integration Constraints & Physical Limits. The derivations for each material technology above (superconductors, high-entropy alloys, metamaterials, carbon-based nanomaterials) are based on the ideal properties of individual materials. In practical engineering systems, integration between materials faces the following unresolved constraints:

Material Compatibility Issues:

- The precise lattice structure required by room-temperature superconductors and the high degree of lattice disorder inherent to high-entropy alloys may produce fundamental conflicts at their interface. If both must operate within the same system (e.g., magnetic confinement coils and structural wall materials of a fusion reactor), interface engineering will become an independent technical bottleneck.

- Room-temperature, ambient-pressure superconductivity itself currently has no reproducibly verified experimental realization. Lattice stability limitations within the BCS theory framework mean this technology may require an entirely new superconducting mechanism. If room-temperature superconductivity at ambient pressure is ultimately proven infeasible, the entire derivation chain dependent on it (lossless transmission, fusion magnetic confinement, electromagnetic shielding, stealth) will require alternatives—possible substitutes include high-temperature superconducting materials (liquid nitrogen temperature range, already commercialized) paired with efficient cryogenic systems, but this increases energy consumption and system complexity.

Physical Limits of Metamaterials:

- **Bandwidth constraint**: The Kramers-Kronig causality relations impose fundamental limits on the frequency response of metamaterials. Perfect broadband cloaking (simultaneously covering all radar bands, infrared, and visible light) may be physically unrealizable. Practically achievable is narrowband cloaking targeting specific detection technology frequency bands, or adaptive metamaterials that switch between different frequency bands.

- **Scale constraint**: The effective feature scale of metamaterials must be comparable to the target wavelength. Structures effective for radar waves (cm-scale) are ineffective for infrared (μm-scale); the two require design layers at entirely different scales.

Inescapable Thermodynamic Constraints:

- Conservation of energy requires that any continuously operating energy system inevitably produces waste heat. Superconducting systems themselves produce no Joule heating, but fusion reactions, biological activity, and computational processes all generate heat. This heat must be discharged to the environment—it can be directionally dissipated or blended with the environmental background, but it cannot be made to disappear.

- Therefore, "stealth" as used in this framework should be understood as "**reducing detectable signatures below the threshold of current or near-future detection technology**," not "physically completely invisible." As human detection technology advances, this threshold will continue to decrease, and stealth difficulty will continue to increase.

4. Electromagnetic & Thermal Signature Control

Integrating the above material technologies, the signature control capabilities of an advanced civilization can be summarized as:

5. High-Precision Energy Manipulation & Long-Range Transmission

If a civilization masters a mature combination of the above material technologies, their energy manipulation precision will far exceed current human levels:

- **Energy transmission efficiency approaching 100%**—Superconductors eliminate transmission losses.
- **Energy conversion at nanoscale**—Nanomaterials enable molecular-level energy conversion.
- **Energy storage density approaching theoretical limits**—Superconducting magnetic energy storage combined with advanced battery technology.
- **Precisely controlled temporal and spatial distribution of energy release**—Quantum computing optimizes energy release strategies.

Long-Range Energy & Information Transmission Networks:
Combining superconductor, quantum-secured communication, and brain-computer interface technologies, advanced civilizations may build a highly covert long-range transmission infrastructure:

Key derivation: It must be noted that the no-communication theorem [28] of quantum mechanics rigorously proves that quantum entanglement alone cannot transmit information—entanglement measurement outcomes appear completely random from the receiver's perspective, requiring a classical channel to complete information transfer. However, if an advanced civilization employs highly directional photon quantum channels (narrow beam, low power, non-broadcast), paired with quantum key distribution (QKD) ensuring unbreakable encryption, such communications would be practically extremely difficult for third parties to detect and decode—**this provides a communication technology explanation for the Fermi Paradox**: it is not that no civilizations are communicating, but that their communication modality possesses extreme directionality and encryption, making it practically difficult for humanity's SETI program to intercept.

Limitations of this explanation: The directed communication hypothesis requires that *all* civilizations in the universe that have achieved communication capability choose directional rather than broadcast communication—this is an AND condition. Only one civilization need choose to broadcast (whether for exploration, display, or carelessness) for SETI programs to theoretically be able to detect it. Therefore, the directed communication hypothesis as a complete explanation for the Fermi Paradox requires at least one additional assumption: (a) some form of interstellar convention or normative pressure drives all civilizations toward concealment; (b) civilizations that choose to broadcast face higher survival risks ("Dark Forest" game-theoretic pressure); or (c) the number of civilizations is extremely small, making the AND condition easier to satisfy. This framework does not presuppose which of the above holds, but merely notes that the directed communication hypothesis as a standalone explanation is incomplete.

Remote operation application derivations:
If both energy and information can be transmitted losslessly to biological bodies over long distances, then:

- **Biological reconnaissance units**—Individuals can be remotely deployed to target areas, continuously receiving energy replenishment and commands, without needing to carry supplies or communication devices.
- **Biological proxy operations**—Central nodes can simultaneously operate multiple remote biological bodies to execute precision tasks (observation, sampling, construction).
- **Unlimited-endurance biological probes**—Biological bodies as probe carriers, achieving theoretically unlimited mission endurance through remote charging.
- **Distributed sensing arrays**—Large numbers of individuals distributed across vast areas, sharing perceptual data, forming a biological-grade distributed radar network.

This explains certain UAP report observations where "biological entities appear to carry no equipment"—equipment may have been integrated within the biological body, or external equipment may simply not be needed.

6. Distributed Civilization Architecture

If all technological thresholds are achieved, advanced civilizations would not need centralized urban structures. Possible organizational forms:

- **Distributed node networks**—Each node possesses complete energy, computing, and life support capabilities.
- **Redundant design**—Loss of any single node does not affect overall civilization operation.

TABLE V. Comprehensive signature control capabilities.

Signature Type	Control Technology	Effect
EM Radiation	Metamaterial cloaking	Significantly reduced detectability in specific bands (Kramers-Kronig bandwidth limited)
Thermal Infrared	Directional thermal radiation + metamaterials	Thermal signature blends with environmental background
Magnetic Field	Superconducting shielding	Internal strong magnetic fields completely contained
Acoustic	Acoustic metamaterials	Active noise cancellation and sound wave redirection
Gravitational	Mass distribution optimization	Reduced local gravitational anomaly signal

TABLE VI. Long-range energy and information transmission network characteristics.

Transmission Type	Technical Pathway	Human Detectability	Theoretical Basis
Energy (short-range)	Inductive / resonance coupled wireless	Very low (near-field)	Existing human prototypes (Qi charging standard)
Energy (long-range)	Directed microwave/laser energy beams	Low (narrow beam)	NASA space solar power research
Energy (superconducting)	Zero-loss superconducting cable	Zero (underground/undersea)	Superconductor zero-resistance property
Information (classical)	Metamaterial waveguide directed transmission	Very low (non-broadcast)	Metamaterial research
Information (quantum-secured)	QKD + highly directional photon channels	Very low (narrow-beam, unbreakable encryption)	Quantum no-cloning theorem

- **Low-signature distribution**—Avoiding detectable signals generated by large-scale concentration.

- **Adaptive reconfiguration**—Nodes can dynamically reconnect and reorganize.

Brain-Computer Networks as Civilization Backbone:

When brain-computer interfaces, long-range energy transmission, and quantum information channels are combined, civilization architecture undergoes a qualitative transformation:

Under this architecture:

- **The civilization has no attackable "capital" or "central hub"**—Every individual is a complete civilization slice.

- **No detectable infrastructure is needed**—No power grid, no communication towers, no transportation network.

- **Biological individuals themselves are the civilization's hardware**—Death is merely a node going offline; knowledge and memories have already been synchronized to the network.

This is entirely opposite to humanity's urbanization trend, but offers overwhelming advantages in stealth and survivability. This also provides an additional corollary: **Advanced civilizations may not construct "buildings" or "cities" as we understand them at all, meaning that subsurface imaging and deep-sea scanning finding no artificial structures cannot rule out the logical possibility of distributed non-material civilization forms—but note that such forms already exceed the boundaries of falsifiable science (see Sec. I E, Epistemological Layering).**

IV. PART III — STRUCTURAL CLASSIFICATION OF EXTRATERRESTRIAL EVENTS

A. Source Classification of Contemporary Extraterrestrial Events

If the existence of highly intelligent civilizations is assumed, then contemporary Earth-related extraterrestrial narratives can be decomposed into the following independent models. Each model has different technological prerequisites and observable characteristics.

1. *Military Misjudgment & Human Technology*

The most probable source of a large proportion of UFO/UAP reports:

- **Misidentification of advanced military vehicles**—Experimental aircraft secretly developed by various nations.

- **Sensor system artifacts**—Known failure modes of radar and infrared sensors.

- **Optical phenomena**—Atmospheric refraction, plasma discharge, ball lightning.

This model requires no extraterrestrial assumptions and should serve as the baseline (null hypothesis) for all analysis.

2. *Sensor Anomalies & Physical Phenomena*

Some UAP reports can be attributed to known but rare physical phenomena:

- **Atmospheric plasma**—Self-sustaining plasma spheres generated by high-energy discharge.

- **Gravitational lensing micro-effects**—Light bending under extreme atmospheric conditions.

TABLE VII. Comparison of traditional and distributed brain-computer civilization architectures.

Traditional Civilization Architecture	Distributed Brain-Computer Civilization Architecture
Cities as nodes	Individuals as nodes
Power grids transmit energy	Directed wireless + underground superconducting networks
EM broadcast communication (easily detectable)	Directional quantum-secured communication (extremely difficult to detect)
Centralized server computation	Biological brain distributed collaborative computation
Physical movement to exchange information	Consciousness-level instant sharing
Single-point failure can paralyze system	Any node can assume global functions

- **Sensor crosstalk**—Signal interference and misidentification between multi-sensor systems.

3. *Extra-Solar System Observer Model*

Assuming civilizations originating from outside the solar system, their interaction with Earth may be extremely indirect. See Part IV, Sec. V B for details.

4. *Intra-Solar System Observation Station & Outpost Model*

Assuming a civilization has established outposts within the solar system. Potential locations with natural shielding conditions include:

- **Lunar lava tubes**—Natural cave structures, closest to Earth.
- **Martian subsurface**—Confirmed presence of water ice and suspected lava tube entrances.
- **Asteroid belt**—Vast in number, dispersed, and difficult to survey individually.
- **Outer planet moons** (Europa, Titan)—Resources such as water ice or liquid hydrocarbons available.

However, these locations share a common characteristic: **environmental conditions are extremely unsuitable for large-scale permanent civilization habitation**—no or extremely thin atmospheres, extreme temperatures, high radiation, and lack of stable ecosystems.

This leads to a key inference: if non-natural facilities exist at these locations, they are far more likely to be **observation stations, relay posts, or automated monitoring outposts** rather than full colonial settlements. The energy and material requirements for maintaining an observation station are far lower than sustaining a colony, which better aligns with the cost logic of interstellar engineering.

a. Existence Spectrum. The presence of non-Earth outposts within the solar system can be arranged along a spectrum:

The harsher the environmental conditions at a given location, the more likely only lower-spectrum facilities (L0–L1) exist there.

b. Multi-Source Cross-Reference Model. Key insight: These observation stations **do not necessarily belong to the same source civilization**. Possible origins include:

- **Forward observation stations of extra-solar civilizations**—Near-field extensions of long-range observation capabilities (cross-references Sec. IV A 3).
- **Monitoring facilities left behind by ancient Martian civilizations**—Automated legacy systems that continue operating after civilizational evacuation (cross-references Sec. V C).
- **Orbital or atmospheric facilities left behind by ancient Venus civilizations**—Observational legacy systems remaining after the runaway greenhouse effect (cross-references Sec. V D).
- **Remote nodes of ancient Earth civilizations**—External monitoring networks established by underground civilizations within the solar system (cross-references Sec. V E).
- **Non-biological automated sentinel systems**—Von Neumann probes or AI-driven independent observation networks (cross-references Sec. IV A 8).

This model positions the intra-solar outpost model as a **cross-referencing hub** within the entire classification framework: a single location within the solar system may simultaneously host facilities from multiple source civilizations, each operating independently with distinct objectives.

It is worth noting that all the above sources represent **directed migration** pathways—civilizations consciously constructing, relocating, and deploying assets. However, the **Panspermia** mechanism discussed in Sec. II A 5 provides a parallel possibility: life may also have been transferred between planets within the solar system through natural mechanisms such as impact ejection and interplanetary drift, without any intentional

TABLE VIII. Existence spectrum for non-Earth outposts within the solar system.

Level	Form	Personnel Requirement	Applicable Environments
L0	Automated sensors / AI sentinels	None	Any location
L1	Unmanned observation station	None (remote-operated)	Lunar lava tubes, asteroid belt
L2	Small relay base	Minimal rotating crew	Martian subsurface, Lagrange points
L3	Limited-personnel outpost	Small permanent staff	Outer planet moons (Europa, Titan)
L4	Full colonial settlement	Large-scale self-sustaining	Only viable on highly habitable bodies

agency. This means that any biological signatures analyzed in the subsequent planetary chapters must account for two source pathways—**directed migration and natural panspermia**—rather than presuming a single mechanism by default.

c. Possible Locations & Corresponding Levels.

- **Lunar lava tubes**—L0–L2. Closest to Earth, ideal for real-time observation of terrestrial activity. Natural caves provide shielding from radiation and micrometeorites.
- **Martian subsurface**—L1–L3. May represent remnant facilities of ancient Martian civilization, or relay stations established by external civilizations utilizing existing subsurface structures.
- **Asteroid belt**—L0–L1. Ideal deployment zone for distributed sensor networks; facilities within a single asteroid are virtually undetectable.
- **Outer planet moons (Europa, Titan)**—L1–L3. Resources such as water ice or liquid hydrocarbons enable long-term self-sustaining outposts.

5. *Ancient Martian Civilization Remnant Model*

Assuming Mars produced a civilization during its early habitable window, which was subsequently destroyed or migrated due to environmental catastrophe. See Part IV, Sec. V C for details.

6. *Ancient Venus Civilization Remnant Model*

Assuming Venus produced life or even civilization during its early habitable window, which was subsequently destroyed or forced to migrate due to a runaway greenhouse effect. Venus's habitable window may have lasted over 2 billion years (approximately 700 million to 3 billion years ago), far exceeding Mars's 500 million to 1 billion years, giving it a unique position in the statistical probability of civilization formation.

Key difference from the Mars model: Venus's runaway greenhouse effect rendered its surface environment (462 °C, 90 atm) [29] completely destructive to all solid-state remnants. However, a habitable zone exists at 50–60 km altitude in Venus's atmosphere where temperature (0 °C to 60 °C) and pressure (∼1 atm) approximate Earth's surface conditions. This provides derivation space for a survival pathway that neither the Earth nor Mars hypotheses can cover—the **atmospheric habitation model**.

See Part IV, Sec. V D for details.

7. *Ancient Earth Civilization Underground Continuation Model*

Assuming an advanced civilization existed on Earth before human civilization and transitioned underground following climate catastrophe or asteroid impact. See Part IV, Sec. V E for details.

8. *Non-Biological Intelligence Probe Model*

Assuming observed phenomena are not direct actions of biological civilizations, but rather:

- **Self-replicating probes (Von Neumann Probes)** [30]—Automated interstellar exploration systems.
- **AI-driven observation networks**—Long-term monitoring systems without biological operators.
- **Dormant sentinel systems**—Pre-positioned devices that activate under specific trigger conditions.

Key characteristic of this model: No assumption of continuous biological entity presence is required.

V. PART IV — CIVILIZATION EXISTENCE MODELS

A. Bayesian Prior Assessment of Sub-Models

Before detailing each existence model, the following table summarizes the prior plausibility, supporting evidence class, primary falsification pathway, and false-positive risk for each sub-model. Priors are qualitative assessments conditioned on current evidence and are subject to revision as new data become available.

TABLE IX. Bayesian prior assessment of civilization existence sub-models.

Sub-Model	Prior Plausibility	Supporting Evidence	Primary Falsification Pathway	False-Positive Risk
Military misjudgment / sensor artifacts	**High**	Documented sensor failure modes; classified aircraft programs	Exhaustive declassification + sensor calibration studies	Low—well-understood mechanisms
Extra-solar observers	**Low–Moderate**	Exoplanet statistics; Drake-type estimates with realistic uncertainty distributions [7]	Comprehensive all-sky technosignature survey yielding null result [31]	Moderate—anthropomorphic projection of observation intent
Ancient Mars habitable environment	**Moderate**	Noachian-era liquid water evidence; atmospheric loss models	Mars sample return showing no biosignatures across multiple sites	Low—habitability ≠ biosphere, clearly bounded claim
Ancient Mars civilization node	**Very Low**	None direct; inferred from habitability window	Comprehensive subsurface radar mapping (SHARAD/MARSIS) + sample return	High—large inferential gap from habitability to technosphere
Ancient Venus habitable environment	**Low–Moderate**	GCM models [32, 33]; but constrained by dry interior evidence [34]	Venus atmospheric entry probe + surface radar (VERITAS/EnVision)	Moderate—model-dependent; recent constraints reduce prior
Ancient Venus civilization / aerial biosphere	**Very Low**	Speculative; phosphine claim unconfirmed [35, 36]; metabolic models challenged [37]	In-situ atmospheric chemistry analysis; cloud-deck sampling	High—multiple unfavorable constraints since 2020
Ancient Earth underground civilization	**Very Low**	Silurian hypothesis framework [38]; no direct evidence	Global subsurface imaging + deep-sea floor survey	Moderate—geological record destruction limits testability
Non-biological intelligence probes	**Low**	Physical feasibility of self-replicating probes [30]	Interstellar object spectral/compositional analysis	Moderate—natural object misidentification
Directed migration (Mars→Venus→Earth relay)	**Very Low**	Temporal ordering of habitability windows; panspermia feasibility	Multi-body biosignature comparison; genetic analysis of any extraterrestrial biology found	High—chain of low-prior assumptions multiplied together
Distributed non-material civilization	**Not assessable**	None; unfalsifiable by design	No empirical pathway exists	N/A—outside scope of falsifiable science

Reading note: "Very Low" prior does not mean "impossible"—it means the claim requires substantially more evidence than currently available before it can compete with simpler explanations.

B. Extra-Solar System Observer Model

1. *Long-Range Observation Civilizations*

If a civilization is located tens to hundreds of light-years away, its observation methods for Earth may include:

- **Spectral analysis**—Identifying biosignatures through atmospheric spectra (e.g., coexistence of oxygen and methane).
- **Radio monitoring**—Receiving radio signals that Earth has been broadcasting since the 1920s.
- **Transit observation**—Analyzing atmospheric composition through Earth transit events.

Such civilizations may already know that life exists on Earth but have not yet initiated direct contact.

2. *Orbital Observation Stations*

If a civilization decides to conduct close-range observation, it may establish observation stations at the following locations:

- **Lagrange points** (L4, L5)—Stable orbits with low long-term maintenance costs.
- **High orbits**—Stable orbits far from Earth's detection range.
- **Asteroid camouflage**—Disguising observation stations as natural asteroids.

3. *Micro Self-Replicating Probes*

Von Neumann Probe [30] model:

- Self-replicating micro-probes can diffuse exponentially across an entire galaxy.
- Assuming each replication cycle takes 1,000 years, covering the entire Milky Way requires only approximately 10^6–10^7 years.
- This is virtually instantaneous on cosmic timescales.

If such probes have already reached the solar system, their size may be too small to be detected by current technology.

4. *Low-Intervention Strategy*

Advanced civilizations may consciously choose not to intervene in Earth's civilizational development (similar to the "Zoo Hypothesis" [39]). Possible reasons include:

- Scientific value of observing natural evolutionary processes.
- Ethical considerations regarding civilizational shock.
- Waiting for Earth civilization to reach a specific technological threshold.

5. *Energy & Stealth Requirements*

Observation stations within the solar system would require:

- **Low-signature energy systems**—Avoiding detection by Earth.
- **Primarily passive observation**—Minimizing active signal emission.
- **Long-term self-sustaining capability**—Potentially requiring thousands of years of unsupplied operation.

C. Ancient Martian Civilization Hypothesis

1. *Early Mars Habitability*

Geological evidence indicates that Mars possessed the following conditions approximately 3.5–4 billion years ago:

- **Liquid water**—Geological evidence of river channels, deltas, and lake sedimentary deposits.
- **Thicker atmosphere**—Atmospheric pressure sufficient to maintain surface liquid water.
- **Magnetic field protection**—Mars once possessed a global magnetic field capable of protecting the atmosphere from solar wind erosion.
- **Suitable temperatures**—Some regions may have had temperatures above freezing.

This habitable window lasted approximately 500 million to 1 billion years—sufficient on Earth's timeline for evolution from single-celled to multi-cellular life.

It must be emphasized that the origin of early Martian life admits two non-mutually-exclusive pathways: independent abiogenesis on Mars itself, or the Panspermia mechanism described in Sec. II A 5—microorganisms from Earth or other bodies arriving on Mars via meteoritic transfer (and vice versa). Distinguishing between these two pathways is critical for the subsequent discussion of civilizational origins: if Martian life originated through natural panspermia, it may share a biochemical foundation with terrestrial life; if it arose independently, it may represent an entirely separate experiment in life.

2. *Civilization Formation Time Window*

Earth took approximately 4 billion years from the origin of life to the emergence of intelligent civilization. Mars's habitable window was approximately 500 million to 1 billion years.

Two possibilities:

1. **Insufficient time**—Martian life did not evolve to the intelligence stage.
2. **Different conditions**—If the Martian environment permitted faster evolutionary rates, or life evolved via different pathways, intelligent life could have emerged in shorter timeframes.

3. *Civilization Extinction or Migration Possibilities*

If Martian civilization once existed, its possible responses to environmental catastrophe (atmospheric loss, disappearance of liquid water):

- **Complete extinction**—Technological level insufficient to cope with environmental change.
- **Underground migration**—Transitioning underground to continue survival.
- **Interplanetary migration**—Migrating to Earth or other celestial bodies.
- **Partial survival**—Small groups enduring in underground or extreme environments.

4. *Underground Shelter Structures*

Structural advantages of Martian underground bases:

- **Radiation shielding**—Mars lacks a global magnetic field and thick atmosphere; underground structures are the only long-term radiation protection.
- **Temperature stability**—Underground temperatures are far more stable than surface temperatures.
- **Water ice presence**—Large quantities of water ice have been confirmed beneath Mars's polar regions and mid-latitudes.
- **Natural caves**—Mars has revealed suspected lava tube entrances that could serve as natural shelter spaces.

5. *Mars–Earth Migration Theory*

If Martian civilization possessed interplanetary travel capabilities, Earth is the nearest habitable target. Derivation of migration pathways:

- The closest Mars–Earth distance is approximately 55 million kilometers.
- With current human rocket technology, this requires approximately 7–9 months.
- Advanced civilization propulsion technology could drastically reduce this travel time.

The migration time window may have been approximately 3–3.5 billion years ago, when Earth already had primitive life but had not yet produced complex organisms.

D. Ancient Venus Civilization Hypothesis

1. *Early Venus Habitability*

NASA GISS climate models (Way et al., 2016 [32]; Way & Del Genio, 2020 [33]) indicate that Venus may have possessed the following conditions approximately 700 million to 3 billion years ago (some models suggest habitable conditions may have begun as early as ∼4.2 billion years ago):

- **Liquid water oceans**—Models project that when early Venus had a slower rotation rate, atmospheric circulation could maintain moderate surface temperatures (approximately 20 °C to 50 °C), sufficient to support stable liquid water bodies.

- **Thick but non-runaway atmosphere**—Early Venus atmospheric CO_2 concentrations may not yet have triggered the positive feedback runaway.

- **Lower solar irradiance**—The early Sun's luminosity was approximately 70–85% of its current level, reducing the initial pressure for greenhouse runaway.

- **Earth-similar volume and gravity**—Venus's mass is 81.5% of Earth's, with surface gravity approximately 90% of Earth's, posing no fundamental constraints on biological evolution.

This habitable window may have lasted over **2 billion years**—far exceeding Mars's 500 million to 1 billion years. By Earth's standards, this timeframe is sufficient to support the complete evolutionary trajectory from single-celled organisms to complex multicellular life.

Parameter Sensitivity Warning: The above habitability window estimate is highly dependent on the initial condition assumptions of the climate model. Key sensitive parameters include:

Some plausible parameter combinations could shorten the habitable window to a few hundred million years or less—in such a scenario, the statistical probability of complex life or civilization forming on Venus would be substantially reduced. This framework's Venus submodel should be interpreted against this backdrop of uncertainty.

Post-2024 Observational Constraints: Since the original Venus habitability models were published, several lines of evidence have tightened the constraints on Venus's habitable past:

- **Dry interior constraint**—Modeling of Venus's interior evolution indicates that Venus's mantle is significantly drier than Earth's, consistent with a long-term arid surface history. This reduces the prior probability that Venus sustained liquid water oceans for billions of years [34].

- **Phosphine re-analysis**—Independent re-analysis of the original ALMA data found no statistically significant phosphine detection [35]. SOFIA observations provided a strict upper limit well below the originally claimed abundance [36].

- **Metabolic model challenges**—Proposed biochemical metabolic pathways for a Venus cloud-layer biosphere have been shown to be inconsistent with observed atmospheric chemistry—the predicted chemical signatures do not match what is actually measured [37].

- **Amino acid stability**—Concentrated sulfuric acid in Venus's cloud layer rapidly destroys amino acid backbones, posing a fundamental challenge to Earth-like biochemistry in the Venus atmosphere [40].

- **Dipeptide stability**—Follow-up studies on peptide bond stability under Venus cloud conditions further constrain the viability of complex organic chemistry in this environment [41].

Balanced assessment: The Venus habitability hypothesis now occupies a more constrained position than in the pre-2024 literature. The early habitability window remains physically possible under specific parameter combinations, but the overall prior has been reduced. This framework retains Venus as a bounded, low-prior branch rather than a co-equal pathway alongside Mars.

As with the Mars chapter (Sec. V C 1), early Venus life may have originated through two pathways: independent local abiogenesis, or natural transfer from other bodies via the Panspermia mechanism (see Sec. II A 5). Notably, because Venus's orbital distance to Earth is shorter (approximately 0.28 AU at closest approach), the efficiency of interplanetary meteoritic transfer may be higher than that of the Mars–Earth pathway, giving panspermia a higher prior probability in the Venus scenario.

2. *Civilization Formation Time Window*

Comparison of Venus and Mars habitable windows:

Venus's longer habitable window implies: if life emerged on Venus, the statistical probability of its evolution to complex life or even intelligent life exceeds that of Mars.

3. *Runaway Greenhouse Effect & Civilization Response*

The mechanism by which Venus lost its habitability—the Runaway Greenhouse Effect—has fundamentally different characteristics from Mars's atmospheric loss:

- **Potentially faster**—Once positive feedback initiates, atmospheric temperature can climb from temperate to hundreds of degrees within millions of years.

- **Greater irreversibility**—Once CO_2 is released from carbonate rocks, there is no reverse deposition mechanism.

- **Complete surface destruction**—The current Venus surface temperature of 462 °C and pressure of 90 atm [29] would completely dissolve any solid-state civilization remnants.

If Venus civilization once existed, its possible responses to the runaway greenhouse effect:

- **Complete extinction**—Technological level insufficient to cope with global atmospheric runaway.

- **Atmospheric migration**—Relocating to the habitable zone in the upper atmosphere (50–60 km altitude).

TABLE X. Parameter sensitivity of the Venus habitability model.

Parameter	Baseline Assumption	If Assumption Changes
Early rotation rate	Slow (similar to present)	Faster rotation → altered circulation, shorter window
Initial water inventory	Earth-like	Less water → oceans may never have formed
Carbon cycle efficiency	Effective carbonate-silicate	Low efficiency → earlier runaway greenhouse
Early solar luminosity	70–85% of present	Relatively stable constraint, lower uncertainty

TABLE XI. Comparison of Venus and Mars habitable windows.

Parameter	Mars	Venus
Habitable window duration	∼500M–1B years	Possibly ∼2B+ years
Habitable period	3.5–4.0 Bya	∼700M–3B years ago
Cause of habitability loss	Magnetic field loss → atmospheric erosion	CO_2 positive feedback → runaway greenhouse
Speed of habitability loss	Gradual (hundreds of millions of years)	Potentially abrupt (millions to tens of millions of years)

- **Interplanetary migration**—Migrating to Earth or Mars.
- **Orbital migration**—Establishing Venus orbital stations or Lagrange point facilities.

4. *Atmospheric Habitation Model*

A unique environmental band exists at 50–60 km altitude in Venus's atmosphere:

This environmental band represents the only **non-solid-surface civilization survival model** within the hypothesis framework.

If a civilization possesses:

- **Buoyant structure engineering**—Exploiting the density difference between Venus's atmospheric composition (primarily CO_2) and breathable gases (N_2/O_2); a nitrogen-oxygen gas mixture is naturally buoyant in Venus's atmosphere. If hydrogen (H_2) or helium (He) is used as the lifting gas instead, buoyancy can increase by approximately 1.5–2×, but hydrogen introduces explosion risk (in oxygen-containing gas mixtures), while helium faces resource acquisition constraints. Practical designs must trade off between buoyancy gain and safety.
- **Sulfuric acid-resistant materials**—Venus's cloud layer contains concentrated sulfuric acid droplets, requiring highly corrosion-resistant hull materials.
- **Closed-loop ecological systems**—Maintaining complete life support cycles within buoyant structures.
- **Atmospheric chemical energy harvesting**—Utilizing chemical gradients in Venus's atmosphere as an energy source.

Then permanent atmospheric habitation does not violate known physical laws.

The unique derivation value of this model: it provides a civilization survival form that neither the Mars nor Earth hypotheses can cover—a **floating civilization**. In this form, a civilization leaves no solid surface remnants whatsoever, rendering it completely invisible in the archaeological sense.

Open Questions on Energy Feasibility:

The atmospheric habitation model faces more stringent energy constraints than underground or undersea models:

This energy dilemma is a key unresolved constraint. If future materials science cannot reduce fusion reactor mass to a range bearable by buoyant structures, the feasibility of atmospheric habitation will depend on whether low-energy-density combinations (chemical gradient + solar + wind) suffice to sustain civilization activity—an open question requiring quantitative analysis.

5. *Venus–Earth Migration Theory*

If Venus civilization possessed interplanetary travel capabilities, Earth is the nearest habitable target:

- The closest Venus–Earth distance is approximately 38 million kilometers (closer than the Mars–Earth minimum of 55 million kilometers).
- The migration time window for Venus civilization may have been approximately 700 million years ago (when the runaway greenhouse effect initiated), at which point Earth already hosted complex multicellular life.
- If migration occurred at an earlier period (1–2 billion years ago), only prokaryotic and simple eukaryotic life existed on Earth.

Potential interweaving of Venus and Mars migration theories:

TABLE XII. Venus atmospheric habitable zone parameters.

Parameter	Value	Comparison with Earth Surface
Temperature	Approx. $-10\,^{\circ}$C to $75\,^{\circ}$C	Comparable (50–55 km most habitable)
Pressure	$\sim$0.2–1 atm	Comparable ($\sim$1 atm at 50 km, $\sim$0.2 atm at 60 km)
Gravity	$8.87\,\mathrm{m/s^2}$ [29]	90% of Earth
Radiation shielding	Dense atmosphere	Superior to Earth

TABLE XIII. Energy feasibility analysis for Venus atmospheric habitation.

Energy Option	Theoretical Feasibility	Engineering Constraints
Atmospheric chemical gradient	Low ($\sim 10^0$–10^1 W/m^2)	Insufficient for civilization-level activity
Solar (above cloud layer)	Medium ($\sim$2600 W/m^2)	Sulfuric acid cloud attenuation
Nuclear fusion	Highest density	Reactor mass vs. buoyancy budget conflict
Wind (super-rotation)	Medium ($\sim$100 m/s)	Requires large structures

- If both Mars and Venus produced civilizations at different epochs, Earth may serve as a common receiving node for two independent migration paths.
- This further strengthens the Part VI derivation of "multi-source model ensembles"—"alien" phenomena on Earth may contain remnants of different planetary origins.

6. *Falsifiability & Observational Constraints*

The Venus hypothesis faces more stringent verification challenges than the Mars hypothesis:

- **Surface remnants are completely inaccessible**—The 462 °C, 90 atm [29] environment limits any lander's lifespan to extremely short durations (the Soviet Venera series survived a maximum of approximately 127 minutes) [42].
- **Geological timescale masking is more thorough**—The Venus surface underwent a global volcanic resurfacing event approximately 300–500 million years ago [43], erasing all earlier geological records.
- **Atmospheric habitation structures are difficult to detect from orbit**—If buoyant structures exist, their scale and characteristics may be indistinguishable from cloud features.

However, the following observation pathways remain viable:

- **Venus atmospheric chemical anomaly analysis**—Searching for non-equilibrium chemical species (e.g., the phosphine controversy, Greaves et al., 2020 [44]).
- **Long-term Venus orbital monitoring**—Searching for artificial structures or regular movement patterns within the atmosphere.
- **Venus surface radar penetration** (e.g., NASA VERITAS [45] [facing budget uncertainty], ESA EnVision [46] [launch targeted for 2031] missions)—Searching for anomalous subsurface structures, though constrained by the high-temperature, high-pressure environment.

E. Ancient Earth Civilization Hypothesis

1. *Geological Timescale Masking Effect*

Over Earth's 4.6-billion-year history, plate tectonics, weathering erosion, and geological cycling would eliminate the vast majority of surface civilization traces within millions of years.

Key question: **If an industrial civilization existed and perished 50 million years ago, what evidence could we find today?**

- Most buildings and metal artifacts would be eroded or buried.
- Plastics and synthetic compounds might leave chemical anomalies in sedimentary layers.
- Large-scale nuclear activity could leave isotope ratio anomalies.
- Sudden carbon emission events in the geological record (e.g., PETM [47]) could be related to industrial activity.

This is known as the **"Silurian Hypothesis"**, formally proposed in 2018 by NASA scientist Gavin Schmidt and astrophysicist Adam Frank [38].

Furthermore, Panspermia (see Sec. II A 5) provides a third possible framework for the origin of terrestrial life: life on Earth may not have arisen entirely from independent local chemistry, but may incorporate biological components delivered from Mars, Venus, or other bodies via natural meteoritic transfer. If this pathway holds,

there exists a deeper biochemical connection between the Martian (Sec. V C) and Venus (Sec. V D) civilization hypotheses discussed in preceding chapters and Earth's own biological history—life on different planets may share a common ancient ancestor encoded in the same genetic language.

2. Advanced Civilization Extinction on Earth

Pathways for advanced civilization extinction:

- **Asteroid impact**—The Cretaceous–Paleogene extinction event 66 million years ago.
- **Supervolcano**—Climate catastrophe caused by massive volcanic activity.
- **Self-destruction**—Nuclear warfare or ecological collapse.
- **Pandemic**—Globally lethal epidemic disease.

3. Underground & Polar Migration

If a civilization foresaw a surface extinction event, possible survival strategies:

- **Underground migration**—Establishing deep underground self-sustaining facilities.
- **Polar migration**—Utilizing polar ice sheets for additional shielding.
- **Oceanic migration**—Leveraging the stability of deep-sea environments.

This model shares the engineering advantages of underground concealment structures with the Antarctic and oceanic models.

4. Antarctic Model

Structural advantages of Antarctica as a base location:

- **Thick ice sheet** (average 2.16 km)—Providing radiation shielding, observational concealment, and environmental isolation.
- **Extremely minimal human activity**—Zero permanent residents across the entire continent; research station personnel fluctuate seasonally.
- **Subglacial lake system**—Such as Lake Vostok (approximately 4 km beneath the ice), capable of providing water resources.
- **Extreme climate**—Average annual temperature of $-57\,^{\circ}$C, naturally reducing human exploration frequency.
- **Limited satellite imaging**—Ice sheet thickness and terrain make underground structures difficult to detect via remote sensing.

Energy derivation: If a civilization possesses controlled nuclear fusion, an Antarctic underground base can sustain energy self-sufficiency long-term, independent of surface supply.

5. Oceanic Subsurface Model

Structural advantages of the ocean floor as a base location:

- **Extreme exploration difficulty in high-pressure deep-sea zones**—Human direct visual observation coverage of the deep seafloor is approximately 0.001% (2025 *Science Advances* study) [48], with exploration of the hadal zone (>6000 m) even more sparse.
- **Excellent sound attenuation**—The acoustic absorption properties of seawater naturally shield internal acoustic activity.
- **Electromagnetic absorption**—Seawater's high absorption rate for electromagnetic waves makes base electromagnetic signatures difficult to detect.
- **Hydrothermal vents**—Can serve as natural energy sources and chemical feedstock origins.

Technical requirements:

- High-pressure resistant structural materials (carbon-based nanocomposites are capable).
- Autonomous energy systems (fusion or geothermal).
- Closed-loop life support systems.

VI. PART V — BASE ENGINEERING & STEALTH MATRIX ANALYSIS

A. Base Engineering & Stealth Matrix

1. Geographic Site Selection

Synthesizing all models, the most probable base site selections follow these principles:

2. Energy Self-Sufficiency

Multi-dimensional analysis matrix for base energy systems:

The combination of fusion + superconducting transmission provides the optimal balance of energy density and stealth.

TABLE XIV. Geographic site selection factors for base placement.

Selection Factor	Optimal Condition	Representative Location
Human activity density	Extremely low	Antarctica, deep sea, desert subsurface
Natural shielding	Thick ice/deep water/rock strata	Antarctic ice sheet, Mariana Trench
Energy accessibility	Geothermal/water resources	Hydrothermal vents, volcanic regions
Detection difficulty	Satellite/sonar/radar blind spots	Sub-ice polar, deep ocean trenches

TABLE XV. Multi-dimensional analysis of base energy systems.

Energy Type	Energy Density	Stealth	Sustainability	Tech Threshold
Compact Fusion	Extremely High	Extremely High	Decades	High
Geothermal Extraction	Medium	High	Continuous	Medium
Hydrothermal Vent	Medium	High	Continuous	Medium
Superconducting Storage	High	Extremely High	Rechargeable	High

3. *Electromagnetic & Thermal Signature Suppression*

Detectable base signatures and suppression strategies:

4. *Gravitational Anomaly Management*

Large underground or undersea structures produce measurable local gravitational anomalies. Management strategies:

- **Mass distribution optimization**—Fitting the base's overall mass distribution to natural geological structures.
- **Low-density material usage**—The low density of carbon-based materials reduces gravitational signals.
- **Distributed architecture**—Avoiding concentrated gravitational anomalies generated by single large structures.
- **Deep site selection**—Natural density variations in the deep crust can mask artificial structural signals.

5. *Material & Structural Advantages*

Comprehensive material-dimension analysis framework:

This framework makes base derivation dependent not only on geographic conditions but on **materials engineering maturity**. The degree of materials science breakthrough directly determines the feasibility, stealth, and sustainability of a base.

VII. PART VII — CROSS-GALACTIC MISSION ARCHITECTURE & HUMAN POSITIONING

A. Cross-Galactic Civilization Deployment Logic

1. *Civilization as Higher-Order Information Structure*

Building upon all preceding derivations, a more expansive framework can be proposed: **The essence of civilization is not buildings, cities, or nations, but a higher-order information structure.**

The persistence of civilization depends on three core conditions:

1. **Stable energy supply**—Sustaining the operation of all physical and computational infrastructure.
2. **Sustainable ecological cycles**—Providing the material basis for biological carriers.
3. **Cross-generational knowledge preservation mechanisms**—Ensuring that information compression and transmission remain uninterrupted.

When any one of these three conditions is destroyed, civilization does not merely "decline"—it undergoes **structural reset to zero**.

2. *Long-Period Model for Cross-Galactic Deployment*

On astronomical timescales, the formation, destruction, and reconstitution of stars and planets are normative processes. The habitability windows of planets within the solar system exhibit a temporal sequence:

This temporal sequence implies a **planetary relay deployment logic**: civilization (or life) can deploy or migrate sequentially among solar system planets as habitability windows shift. Mars lost habitability first, Venus second, and Earth is currently the only node still within its habitable window.

TABLE XVI. Base signature suppression strategies.

Signature	Source	Suppression Method
EM Radiation	Communications/energy systems	Metamaterial cloaking + superconducting shielding
Thermal Emission	Energy system waste heat	Directional heat dissipation + environmental blending
Acoustic Vibration	Mechanical systems	Acoustic metamaterials + structural isolation
Gravitational Anomaly	Massive structures	Mass distribution optimization

TABLE XVII. Comprehensive material-dimension analysis framework.

Dimension	Key Variables
Energy	Fusion + superconducting transmission
Materials	HEAs / metamaterials / topological materials / carbon-based nanomaterials
Stealth	EM redirection / thermal signature suppression / acoustic shielding
Sustainability	Closed ecology + material self-repair
Computing	Topological quantum computation

If civilization is understood as a cross-galactic deployment project, its construction timeline spans tens of thousands to millions of years. This is not a single-generation plan but a long-term project encompassing an entire species.

Staged model for cross-galactic deployment:

During the interplanetary expansion stage—the phase that Martian and Earth civilizations may have experienced within this hypothesis—any planetary-scale natural catastrophe could sever the entire supply chain.

3. *Natural Panspermia & Directed Migration Dual-Track Model*

The diffusion of life across the universe proceeds along two parallel tracks:

Track One: Natural Panspermia

- Microorganisms migrate between planets via meteorite transport.
- Requires no technology and no intent.
- Depends on the statistical probability of impact events.
- On cosmic-lifetime timescales, this is a high-probability process.

Track Two: Directed Migration

- Advanced species actively establish cross-planetary nodes.
- Depends on energy control, closed ecological systems, and manufacturing capability.
- Requires thousands to millions of years of sustained construction.
- Extremely vulnerable prior to construction completion.

Interweaving of the two tracks:

If early Mars possessed a habitable environment, life may have exchanged naturally between Mars and Earth (Track One). Civilization may then have conducted deliberate artificial deployment on the foundation of this natural dispersal (Track Two).

This implies that life on Earth may possess two simultaneous layers of origin:

- **Biological level**: A product of natural panspermia.
- **Civilizational level**: A node within a cross-galactic deployment project.

Distinction Criteria for the Two Tracks:

If biology or biological fossils are discovered on Mars, Venus, or other celestial bodies in the future, distinguishing natural panspermia from directed migration will become a core scientific question. Possible criteria include:

This criteria table is only a preliminary framework; actual distinction may require cross-comparison of multiple criteria. No single criterion is sufficient to definitively distinguish between the two pathways.

B. Planetary-Level Severance & Civilization Reset Cycles

1. *Cosmic-Scale Severance Mechanisms*

When cross-galactic deployment remains incomplete, the following cosmic-scale events can trigger total collapse of the civilization network:

The common characteristic of these events: **destruction of the ecological supply chain and information carriers.**

TABLE XVIII. Solar system planetary habitability windows.

Planet	Habitable Window	Duration	Mechanism of Habitability Loss
Mars	∼3.5–4.0 Bya	∼500M–1B years	Magnetic field loss → atmospheric erosion
Venus	∼700M–3B years ago	Possibly ∼2B+ years	CO_2 positive feedback → runaway greenhouse
Earth	∼3.8B years ago to present	∼3.8B years (ongoing)	Future solar luminosity increase will terminate habitability

TABLE XIX. Staged model for cross-galactic deployment.

Stage	Timescale	Core Objective	Risk Level
Single-planet civilization	0–10^4 years	Establish planetary-scale energy control	Extremely high (single point of fai
Interplanetary expansion	10^4–10^6 years	Establish adjacent planetary nodes	High (fragile supply chain)
Intra-stellar-system network	10^6–10^7 years	Complete multi-node redundancy within the stellar system	Medium (partial node self-sufficien
Cross-galactic diffusion	10^7–10^9 years	Deploy to neighboring stellar systems	Low (multi-system redundancy)

2. *Structural Model of Civilization Reset*

When a planetary-scale catastrophe occurs, civilizational collapse follows a specific sequence:

1. **First-layer collapse: Communication and navigation disruption**—Cross-planetary nodes lose contact.
2. **Second-layer collapse: Energy network severance**—Nodes dependent on external supply lose power.
3. **Third-layer collapse: Loss of manufacturing capability**—Critical infrastructure can no longer be repaired or replaced.
4. **Fourth-layer collapse: Knowledge transmission rupture**—High-dimensional knowledge loses its carriers; only oral fragments persist.
5. **Terminal outcome: Civilization reverts to a low-complexity state**—Only genetic-level continuity and mythological memory remain.

This cycle of "repeatedly being knocked back to the primitive state" is a predictable phenomenon on cosmic timescales.

3. *Supply Chain Severance & Memory Compression*

After civilizational severance, information survival follows a hierarchical structure:

Key observation: Myths across globally distinct cultures exhibit striking structural similarities—flood narratives, descending sky gods, memories of lost civilizations. These may not be mere cultural coincidence but rather **compressed encodings of the same civilizational severance event**, independently preserved as fragments among different surviving populations.

Human DNA also contains incompletely understood "genetic bottleneck" events (e.g., the Toba supervolcano eruption approximately 74,000 years ago and the contemporaneous genetic bottleneck event [53] (the causal link between the two remains debated; the bottleneck may instead stem from founder effects during the Out-of-Africa migration)). These bottlenecks may mark temporal nodes of civilization reset.

Competing Hypotheses Statement: The structural similarity of global myths admits multiple non-mutually-exclusive explanatory models, and this framework does not reject any of them:

1. **Cognitive universality hypothesis** (modern extension of Jungian archetype theory)—the shared neural architecture of the human brain naturally produces similar narrative structures.
2. **Common natural experience hypothesis**—global natural disasters such as floods, volcanoes, and earthquakes produce parallel narratives in independent cultures.
3. **Cultural diffusion hypothesis**—cultural contact along early human migration routes propagated core narrative motifs.
4. **Civilization severance encoding hypothesis** (proposed by this framework)—structural similarities reflect compressed memories of the same one or more civilization severance events.

The above hypotheses carry equal prior weight at the current level of evidence. This framework proposes the fourth hypothesis as a supplement, not a replacement for the first three. Distinguishing among these hypotheses requires evidence independent of mythology itself (e.g., contemporaneous anomalies in the geological record, non-natural genetic markers in ancient DNA, etc.).

4. *Mars Phase: Possible Early Deployment Node*

Re-examining the Mars civilization hypothesis from Part IV, Sec. V C within the cross-galactic deployment framework:

TABLE XX. Distinction criteria for natural panspermia vs. directed migration.

Criterion	Natural Panspermia Expected	Directed Migration Expected
Genetic code	Shared biochemical systems (if homologous)	Possible artificial gene editing traces
Isotope ratios	Natural nucleosynthesis patterns	Possible non-natural ratios
Biochemical complexity	Evolutionary gradient	Sudden high-complexity appearance
Spatial distribution	Random / impact debris trajectories	Non-random, concentrated at habitable sites
Accompanying materials	Natural rock matrix	Possible non-natural materials

TABLE XXI. Cosmic-scale severance mechanisms.

Severance Type	Mechanism	Impact Radius	Historical Evidence
Gamma-ray burst (GRB)	High-energy radiation released by nearby stellar collapse	Thousands of light-years	Possible link to the Ordovician mass extinction [49]
Extreme solar storm	Stellar superflare	Entire stellar system	A thousandfold version of the Carrington Event (1859) [50]
Planetary chain impact	Asteroid belt perturbation triggering cascading impacts	Planetary scale	Hypothesized Late Heavy Bombardment (currently debated) [51]
Severe orbital shift	Gravitational perturbation from nearby stars	Stellar system	Long-period orbital instabilities
Supernova explosion	Nearby massive star detonation	Tens of light-years	Pliocene seabed iron-60 deposits [52]

- **The Martian habitability window (3.5–4.0 billion years ago) may represent the first deployment attempt.**
- The choices facing Martian civilization (if it existed) in the face of environmental upheaval:
  - Constructing enclosed dome structures
  - Transferring deployment to Earth
  - Establishing multi-node redundancy backups
- **Earth may have served as the backup node following the failure of the Mars node.**

However, if cross-galactic communication and energy networks were severed during the deployment process, the Earth node may never have completed initialization—leaving behind only biological-level seeding rather than a complete civilizational transfer.

5. *Venus Phase: Possible Mid-Period Deployment Node*

Re-examining the Venus civilization hypothesis from Part IV, Sec. V D within the cross-galactic deployment framework:

- **Venus's habitable window (700 million to 3 billion years ago) falls chronologically after Mars (3.5–4 billion years ago) but before the emergence of Earth civilization.** This positions Venus as a potential relay node between Mars and Earth.
- The choices facing Venus civilization (if it existed) in the face of the runaway greenhouse effect:
  - Migrating to the upper atmospheric habitable zone (atmospheric habitation model)
  - Transferring deployment to Earth
  - Establishing Venus orbital facilities as observation and relay stations
- **Earth may simultaneously serve as the common backup target following the failure of both the Mars and Venus nodes.**

The Venus phase adds a key derivation to the deployment framework:

- If Mars civilization migrated to Venus 3.5 billion years ago (when Venus was still within its habitable window), then Venus may represent the first backup node for Martian civilization.
- When Venus's habitable window also closed, civilization faced a secondary migration, ultimately directed toward Earth.
- This forms a **Mars → Venus → Earth** three-stage deployment pathway, or alternatively a **dual-source convergence model** where Mars and Venus each migrated to Earth independently.

6. *Earth Phase: Biological Continuity & Civilization Rebuild*

If Earth is indeed a receiving node from a cross-galactic deployment, then human history may not be an evolutionary story "starting from zero" but rather **a reboot following one (or multiple) incomplete deployments**.

Observations supporting this inference:

- **The anomalous acceleration of human civilization**: From the agricultural revolution to space travel in approximately 12,000 years—on cosmic timescales, this is virtually explosive.
- **Genetic bottleneck events**: Multiple episodes in human history where the population crashed to extremely small numbers.
- **Structural similarities in cross-cultural mythology**: Themes of "lost civilizations,"

TABLE XXII. Information survival hierarchy after civilizational severance.

Survival Layer	Carrier	Fidelity	Persistence
Genetic memory	DNA sequences	High (but not consciously accessible)	Millions of years
Instinctive behavior	Neural circuits	Medium	Tens of thousands of years
Myth and religion	Oral tradition	Low (highly compressed and distorted)	Thousands of years
Written records	Writing systems	Medium-high	Hundreds to thousands of years
Digital data	Electronic media	Extremely high (but carrier is fragile)	Tens to hundreds of years

"departing gods," and "great floods" appear independently across the globe.

- **Rapid stepwise expansion of cognitive capacity**: Human brain volume approximately tripled over ∼2 million years [54] (*Homo habilis* → *Homo sapiens*). Mainstream evolutionary explanations (social intelligence hypothesis, language co-evolution, cultural ratchet effect, etc.) already provide multi-factor models. However, within the cross-galactic deployment framework, an additional non-mutually-exclusive speculation is: if Earth received external biological seeding, the introduced genetic material may have already contained gene combinations with high cognitive potential, making the substrate of natural selection more favorable for rapid cognitive expansion. This speculation does not replace evolutionary explanations but is superimposed upon them.

C. Human Mission Positioning

1. *Paradigm Shift from Contingency to Necessity*

In the conventional narrative, human civilization is a contingent product of Earth. Within the cross-galactic deployment framework, the emergence of human civilization may represent **the natural continuation of an incomplete engineering project**.

If so, the core mission of the current phase is not merely economic development or national competition, but a structurally significant civilizational leap:

2. *Multi-Node Civilization Building*

3. *Critical Conditions for Civilization Survival*

When civilization achieves the following conditions, planetary-scale catastrophe will no longer equate to civilizational reset:

1. **At least two planetary nodes possess full self-sufficiency**—Energy, ecology, and manufacturing all operate independently.
2. **Cross-node knowledge synchronization mechanisms are established**—The knowledge base of any single node can be fully restored at other nodes.
3. **Inter-node energy and information transmission is independent of any single planet**—No reliance on the infrastructure of any one planet.

Prior to achieving these conditions, human civilization remains a **high-risk single-point system**.

4. *Structural Conclusion*

From the cosmic-scale perspective:

- **Single-planet civilization is a transitional phase**—Not a steady state, but a countdown awaiting the next reset.
- **Multi-node civilization possesses long-term stability**—Redundant design ensures that the loss of any single node is non-fatal.
- **Cross-galactic civilization is the natural outcome given sufficient time**—Not a miracle, but the inevitable product of survivorship bias.

Whether humanity has ever completed this step cannot be verified. But if the goal is to avoid being reset once more, the only viable path forward is:

To evolve from an Earth-bound civilization into a cross-galactic civilization.

The mission is clear. The window of opportunity is finite.

VIII. PART VIII — VERIFIABLE OBSERVATION PATHWAYS

A. Empirical Strategy

The value of this hypothesis lies in its falsifiability. The following are specific observation and experimental pathways:

1. *Instrument Resolution & Detection Thresholds*

Before evaluating specific observation pathways, it is essential to establish the current detection floor for each

TABLE XXIII. Multi-node civilization building: missions and progress.

Mission	Objective	Current Progress
Establish multi-planetary survival capability	Self-sustaining colonies on Mars/Moon	Early exploration phase (SpaceX [3], Artemis [4])
Enhance planetary-scale risk resilience	Asteroid defense + climate stabilization	Initial technology validation (DART mission [55])
Establish cross-generational information preservation systems	Knowledge storage on ten-thousand-year timescales	Construction phase (Long Now Foundation's 10,000-Year Clock [5] under physical construction in Texas)
Transition civilization from single-point to multi-node	Distributed civilization architecture	Theoretical phase

method. The table below maps each sub-model's minimum detectable signature against the resolution limits of existing instruments. This determines whether a negative result constitutes a meaningful constraint or merely reflects insufficient instrument sensitivity.

Key implication: The claim "if we find nothing, the sub-model is weakened" must be qualified by instrument capability. For sub-models where current instruments lack the resolution to detect the predicted signature, a null result is uninformative rather than constraining. This table should be updated as new missions become operational.

2. *Subsurface Imaging*

- **High-resolution seismic wave tomography**—Using natural earthquakes and artificial sources to build 3D crustal density maps.
- **Muon Tomography** [57]—Leveraging the rock-penetrating properties of cosmic ray muons to detect underground cavities.
- **Gravity gradient measurements**—Using satellite gravity data (e.g., GRACE/GRACE-FO [56]) to search for anomalous density distributions.

Determination criterion: If high-resolution subsurface imaging fails to detect artificial structural features in candidate regions, then relevant underground base models can be weakened.

3. *Deep-Sea Scanning*

- **Comprehensive ocean floor topographic scanning**—As of 2025, global high-resolution seabed mapping coverage is approximately 27% (Seabed 2030 project [58] data), with over 70% of the ocean floor remaining unmapped.
- **Deep-sea sonar anomaly analysis**—Searching for non-natural echo patterns.
- **Deep-sea thermal anomaly monitoring**—Local temperature anomalies exceeding natural geothermal models.

The ongoing Seabed 2030 project [58] will continue to expand global seabed mapping coverage. **Determination criterion**: If comprehensive deep-sea scanning reveals no artificial structural features, then oceanic base models can be eliminated.

4. *Gravitational Anomaly Monitoring*

- **High-precision local gravity measurements**—Deploying high-sensitivity gravimeters in candidate regions.
- **Gravitational tidal residual analysis**—Searching for gravitational variations unexplainable by natural geological models.
- **Satellite gravity field time series**—Monitoring non-natural temporal variation patterns in gravity fields.

Determination criterion: If gravitational anomaly monitoring fails over the long term to identify non-natural distribution patterns, then base energy models require revision.

5. *Electromagnetic Spectrum Monitoring*

- **Extremely Low Frequency (ELF) monitoring**—Searching for artificial electromagnetic signals originating from underground or undersea sources.
- **Non-natural spectral pattern searches**—Analyzing electromagnetic background noise for artificial modulation signatures.
- **Quantum field fluctuation monitoring**—Searching for anomalous fluctuations exceeding natural quantum noise.

Detectability context: Sheikh et al. (2025) [31] demonstrated that Earth-level technosignatures span 13 orders of magnitude in detectability depending on the emission mode, with certain radio-frequency modes being far more conspicuous than non-radio signatures. This implies that the hypothesis of universal deliberate concealment (directed encrypted communication resolving the Fermi Paradox) requires a more rigorous selection model than intuition alone: even a single civilization choosing broadcast mode would be potentially detectable across significant interstellar distances.

6. *Anomalous Material Sample Search*

- **Anomalous isotope ratios in meteorite and geological samples**—Searching for products of non-natural nucleosynthesis processes.

TABLE XXIV. Instrument resolution and detection thresholds for each observation pathway.

Detection Method	Target Sub-Model	Minimum Signature Required	Current Instrument Capability	Constraint Strength of Null Result
Satellite gravimetry (GRACE-FO [56])	Underground base (mass anomaly)	$\sim 10^{10}$ kg localized mass concentration	Effective spatial resolution ~300 km; sensitive to large-scale mass redistribution	**Weak** for structures <100 km; strong for continent-scale anomalies
Muon tomography [57]	Underground cavity	Density contrast >5% over ~100 m scale	Penetration depth ~1 km in rock; ~10 m resolution at shallow depths	**Strong** for shallow (<1 km), localized targets; negligible for deep or distributed structures
Seabed multibeam sonar (Seabed 2030 [58])	Undersea base	Geometric anomaly >100 m on seafloor	27.3% of ocean floor mapped to modern standards (as of 2025); resolution ~100 m at abyssal depths	**Weak** globally (73% unmapped); strong in surveyed regions
Deep-sea visual observation [48]	Undersea artificial structures	Visual-scale artificial features	<0.001% of deep seafloor directly observed	**Very weak**—near-zero global coverage
Mars subsurface radar (SHARAD [59] / MARSIS [60])	Mars subsurface structures	Dielectric contrast at >10 m scale	SHARAD: ~15 m vertical resolution, ~1 km penetration; MARSIS: ~150 m vertical, ~5 km penetration	**Moderate** for large cavities; weak for small or deep structures
Venus SAR (VERITAS [45] / EnVision [46])	Venus subsurface anomalies	Surface/near-surface geometric anomaly	VERITAS: ~30 m resolution (status uncertain); EnVision VenSAR: ~30 m (launch ~2031)	**Pending**—instruments not yet operational
Technosignature survey [31]	Extra-solar observers, directed EM	Earth-level radio/optical leakage at interstellar distances	Detectable across 13 orders of magnitude depending on mode; strongest radio modes detectable to ~100 pc	**Moderate** for nearby sources; weak for distant or deliberately quiet civilizations
Atmospheric chemistry probes	Venus aerial biosphere	Non-equilibrium chemical species (ppb-level)	No long-duration Venus atmospheric probe currently operational	**Pending**—no in-situ data available

- **Anomalous alloys or material structures**—Searching geological records for material combinations that should not exist naturally.

- **Nanostructure anomalies**—Searching for nanoscale structures with artificially designed features.

7. *Mars & Lunar Subsurface Exploration*

- **Mars subsurface radar detection** (e.g., SHARAD [59], MARSIS [60])—Searching for artificial cavities or structures beneath the Martian surface.

- **Lunar lava tube detection**—Using lunar orbiter radar and gravity data to search for anomalous cavities.

- **Sample return missions**—Retrieving samples from Martian or lunar subsurface to analyze for artificial material traces.

Determination criterion: If Mars and lunar subsurface exploration reveals no artificial structures, then relevant base models can be weakened.

8. *Interstellar Object Observation*

- **Interstellar object spectral and orbital analysis**—Conducting spectral, orbital, and non-gravitational acceleration analysis of interstellar objects entering the solar system (e.g., 1I/'Oumuamua [61], 2I/Borisov [62], 3I/ATLAS [63]), searching for non-natural signatures.

- **Vera Rubin Observatory** [64]—The Rubin Observatory has begun scientific operations, issuing its first batch of scientific alerts on 2026-02-25 [65]. Its Legacy Survey of Space and Time (LSST) is expected to detect dozens of interstellar objects per year, dramatically expanding the sample size and enabling statistical characterization of the interstellar object population.

- **Interstellar object intercept missions**—Rapid intercept mission concepts such as ESA Comet Interceptor [66] could enable close-range observation or even sampling of interstellar objects.

The growing sample of interstellar objects—from a single anomalous case (1I/'Oumuamua) to a population of three with distinct characteristics—strengthens the empirical foundation for distinguishing natural panspermia vehicles from potential artificial probes. Population-level statistical analysis becomes feasible as the sample grows.

Determination criterion: If large samples of interstellar objects all exhibit spectral and orbital characteristics consistent with natural bodies, then the prior probability of the Von Neumann probe model (Sec. IV A 8) should be adjusted downward. Conversely, if any interstellar object exhibits features unexplainable by natural processes (anomalous albedo, non-gravitational acceleration without cometary activity, anomalous shape, etc.), further investigation is warranted.

9. *Venus Atmospheric & Surface Exploration*

- **Venus atmospheric chemical anomaly analysis**—Searching for non-equilibrium chemical species in the atmosphere, particularly molecules that should not stably exist under Venus atmospheric conditions (e.g., phosphine, Greaves et al., 2020 [44]).

- **Long-term Venus upper atmosphere monitoring**—Using orbiters to search for anomalous structures, regular movement patterns, or non-natural reflective features at 50–60 km altitude.

- **Venus surface radar penetration imaging**—Utilizing synthetic aperture radar (e.g., NASA VERITAS mission [45] X-band SAR [planned, facing budget uncertainty], ESA EnVision mission [46] VenSAR [entered construction phase, launch targeted for 2031]) to penetrate cloud cover and search for possible anomalous subsurface structures.

- **Venus atmospheric entry probes**—Deploying buoyant probes capable of long-duration residence in the upper atmosphere for direct sampling and analysis of atmospheric composition and particulate structures.

Determination criterion: If Venus atmospheric chemical analysis reveals no non-natural compounds and long-term upper atmosphere monitoring detects no artificial structural features, then the Venus atmospheric habitation model can be weakened. If Venus surface radar penetration finds no artificial structural traces beneath the global resurfacing layer [43], then the ancient Venus surface civilization model can be further weakened.

IX. STRUCTURAL SUMMARY

If advanced civilizations exist, their key technological breakthroughs will concentrate on:

1. **Maximizing energy density**—Compact fusion frees bases from external energy dependence.
2. **Zero-loss energy transmission**—Room-temperature superconductors eliminate energy loss during transmission.
3. **Complete material adaptation to extreme environments**—High-entropy alloys and carbon-based nanomaterials make deep-sea and polar construction feasible.
4. **Controllable electromagnetic and thermal signatures**—Metamaterials and superconducting shielding achieve physical-level stealth.
5. **Computing capability transcending classical limits**—Topological quantum computation makes complex system simulation and optimization possible.
6. **Cross-galactic deployment architecture**—Transitioning civilization from a single-point system to multi-node redundancy, ensuring that planetary-scale catastrophe no longer equates to civilizational reset.
7. **Planetary relay deployment**—The temporal sequence of habitability windows among solar system planets (Mars → Venus → Earth) provides civilization with a natural relay pathway for interplanetary migration; Venus's extended habitable window and atmospheric habitation potential expand the diversity of civilization survival forms.

These derivations are built upon extensions of known physical theories, not violations of fundamental laws.

Directions for further development:

- Gravitational manipulation material hypothesis
- Negative energy density and vacuum engineering
- Macroscopic quantum field stabilization technology
- Physical implementation of cross-generational knowledge preservation mechanisms
- Materials and energy derivations for Venus atmospheric habitation engineering

The overall model progressively shifts from narrative-level to engineering- and physics-derivable structures.

This hypothesis framework is designed to be progressively filled with rigorous derivations, evidence references, and falsifiable predictions.

Appendix A: Cognitive Deconstruction of the "Alien" Concept

Note: This appendix presents a cultural-analogy framework for the semantic decomposition of the term "alien." It is positioned as supplementary context rather than part of the core evidential chain. Part III (Sec. IV) provides the structural analysis; this appendix explores the cognitive and linguistic dimensions.

1. Limitations of the Single-Species Assumption

The popular cultural concept of "aliens" contains an implicit fundamental assumption: **A single extraterrestrial species is interacting with humanity.**

This assumption is extremely unlikely to hold, for the following reasons:

- Conditions for life formation arise independently across different times and spaces in the universe.
- Different civilizations may have radically different evolutionary paths, technological levels, and biological morphologies.
- So-called "alien events" may have entirely different sources.

2. Multi-Source Model Ensemble

"Aliens" may be the projected ensemble of the following entities:

1. **Interplanetary civilization observers**—Organic life from outside the solar system.
2. **Ancient Earth civilization remnants**—Descendants of Earth's intelligent life predating human civilization.
3. **Inorganic intelligence systems**—Self-replicating probes or AI-driven observation networks.
4. **Biological-mechanical hybrids**—Deep fusion of organic life and mechanical systems.
5. **Martian migration descendants**—Continuation of ancient Martian civilization migrated to Earth.
6. **Venus migration descendants**—Continuation of ancient Venus civilization migrated to Earth following the runaway greenhouse effect.

In the human context, these radically different entities are compressed into the single term "alien."

3. Civilization Morphological Diversity

Different origin pathways will produce radically different civilization morphologies:

4. Human Cognitive Projection

Humans tend to project unknown phenomena into single narratives—this is a natural result of cognitive simplification:

- **Anthropomorphization tendency**—Imagining extraterrestrial intelligence in human-like forms.
- **Single-source assumption**—Attributing all unidentified phenomena to the same origin.
- **Intent projection**—Assuming extraterrestrial behavior has humanly comprehensible purposes.
- **Cultural framework filtering**—Different cultures interpret the same phenomenon in entirely different ways.

Recognizing that "alien" is a semantic compression concept, not a definitive species description, is a necessary prerequisite for rational analysis.

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Data and Code Availability: The complete source files, Markdown versions (English and Traditional Chinese), and independent verification report are available at https://github.com/djchrisssssss/cosmic-diaspora-hypothesis.

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[1] ITER Organization, ITER — international thermonuclear experimental reactor, https://www.iter.org/, accessed 2026.
[2] IBM, IBM quantum hardware & roadmap, https://www.ibm.com/quantum/hardware, accessed 2026.
[3] SpaceX, Human spaceflight: Mars, https://www.spacex.com/human-spaceflight/mars, accessed 2026.
[4] NASA, Artemis program, https://www.nasa.gov/humans-in-space/artemis/ (), accessed 2026.
[5] Long Now Foundation, The 10,000 year clock, https://longnow.org/clock/, accessed 2026.
[6] N. Bostrom, *Anthropic Bias: Observation Selection Effects in Science and Philosophy* (Routledge, 2002).
[7] A. Sandberg, E. Drexler, and T. Ord, Dissolving the Fermi paradox (2018), arXiv:1806.02404, arXiv:1806.02404 [physics.pop-ph].
[8] J. L. Green, S. Byrne, S. Cofield, L. Hays, T. Hoehler, M. Neveu, M. Parenteau, M. Race, B. Robles, and B. Smith, Call for a framework for reporting evidence for life beyond Earth, Nature **598**, 575 (2021).
[9] L. Chomiuk and M. S. Povich, Toward a unification of star formation rate determinations in the Milky Way and other galaxies, The Astronomical Journal **142**, 197 (2011).
[10] NASA, Kepler space telescope mission, https://science.nasa.gov/mission/kepler/ (), accessed 2026.
[11] NASA, TESS (transiting exoplanet survey satellite) mission, https://science.nasa.gov/mission/tess/ (), accessed 2026.
[12] E. A. Petigura, A. W. Howard, and G. W. Marcy, Prevalence of earth-size planets orbiting sun-like stars, Proceedings of the National Academy of Sciences **110**, 19273 (2013).
[13] B. A. McGuire, 2021 census of interstellar, circumstellar, extragalactic, protoplanetary disk, and exoplanetary molecules, The Astrophysical Journal Supplement Series **259**, 30 (2022).
[14] V. M. Rivilla *et al.*, Interstellar glycine detected toward a solar-type protostar IRAS 16293-2422b (2025), research Square preprint.
[15] D. P. Glavin, J. E. Elsila, H. L. McLain, *et al.*, Extraterrestrial amino acids and L-enantiomeric excesses in the CM2 carbonaceous chondrites Aguas Zarcas and Murchison, Meteoritics & Planetary Science **56**, 148 (2021).
[16] K. Takai *et al.*, Cell proliferation at 122 °C and isotopically heavy CH_4 production by a hyperthermophilic methanogen under high-pressure cultivation, Proceedings of the National Academy of Sciences **105**, 10949 (2008).
[17] D. Slade and M. Radman, Oxidative stress resistance in *Deinococcus radiodurans*, Microbiology and Molecular Biology Reviews **75**, 133 (2011).
[18] R. J. Worth, S. Sigurdsson, and C. H. House, Seeding life on the moons of the outer planets via lithopanspermia, Astrobiology **13**, 1155 (2013).
[19] M. Lingam, A. Balbi, and S. M. Mahajan, A Bayesian analysis of the probability of the origin of life per site conducive to abiogenesis, Astrobiology 10.1089/ast.2023.0155 (2024), pubMed 39159441.
[20] N. S. Kardashev, Transmission of information by extraterrestrial civilizations, Soviet Astronomy **8**, 217 (1964).
[21] W. Meissner and R. Ochsenfeld, Ein neuer Effekt bei Eintritt der Supraleitfähigkeit, Naturwissenschaften **21**, 787 (1933).
[22] H. Essen and M. C. N. Fiolhais, Meissner effect, diamagnetism, and classical physics — a review, American

TABLE XXV. Civilization morphological diversity by origin pathway.

Origin	Possible Form	Technological Signature	Interaction Mode
Extra-solar	Entirely heterogeneous biology	Interstellar travel capability	Extremely low in
Ancient Earth civilization	Possibly biologically related to humans	Underground engineering technology	Possible intermit
AI probes	Non-biological	Self-replication capability	Passive observati
Martian migration	Biology adapted to Martian environment	Interplanetary technology	May have integra
Venus migration	Possibly adapted to atmospheric habitation	Buoyant engineering / interplanetary technology	May have integra

Journal of Physics **80**, 164 (2012).

[23] D. B. Miracle and O. N. Senkov, A critical review of high entropy alloys and related concepts, Acta Materialia **122**, 448 (2017).

[24] M. Z. Hasan and C. L. Kane, Colloquium: Topological insulators, Reviews of Modern Physics **82**, 3045 (2010).

[25] J. B. Pendry, D. Schurig, and D. R. Smith, Controlling electromagnetic fields, Science **312**, 1780 (2006).

[26] C. Lee, X. Wei, J. W. Kysar, and J. Hone, Measurement of the elastic properties and intrinsic strength of monolayer graphene, Science **321**, 385 (2008).

[27] M.-F. Yu, O. Lourie, M. J. Dyer, K. Moloni, T. F. Kelly, and R. S. Ruoff, Strength and breaking mechanism of multiwalled carbon nanotubes under tensile load, Science **287**, 637 (2000).

[28] A. Peres and D. R. Terno, Quantum information and relativity theory, Reviews of Modern Physics **76**, 93 (2004).

[29] D. R. Williams, Venus fact sheet, `https://nssdc.gsfc.nasa.gov/planetary/factsheet/venusfact.html` (2024), NASA Goddard Space Flight Center, Accessed 2026.

[30] R. A. Freitas, Jr., A self-reproducing interstellar probe, Journal of the British Interplanetary Society **33**, 251 (1980).

[31] S. Z. Sheikh *et al.*, Earth detecting Earth: At what distance could Earth's constellation of technosignatures be detected with present-day technology? (2025), arXiv:2502.02614, arXiv:2502.02614 [astro-ph.EP].

[32] M. J. Way, A. D. Del Genio, N. Y. Kiang, L. E. Sohl, D. H. Grinspoon, I. Aleinov, M. Kelley, and T. Clune, Was Venus the first habitable world of our solar system?, Geophysical Research Letters **43**, 8376 (2016).

[33] M. J. Way and A. D. Del Genio, Venusian habitable climate scenarios: Modeling Venus through time and applications to slowly rotating Venus-like exoplanets, Journal of Geophysical Research: Planets **125**, e2019JE006276 (2020).

[34] T. Constantinou, O. Shorttle, and E. Maynes, Dry interior constraints on the habitability of Venus, Nature Astronomy (2024), modeling of Venus interior evolution indicating significantly drier mantle than Earth.

[35] I. A. G. Snellen, L. Guzman-Ramirez, M. R. Hogerheijde, A. P. S. Hygate, and F. F. S. van der Tak, Re-analysis of the 267 GHz ALMA observations of Venus: No statistically significant detection of phosphine, Astronomy & Astrophysics **644**, L2 (2020).

[36] M. A. Cordiner, G. L. Villanueva, H. Wiesemeyer, S. N. Milam, M. de Val-Borro, C. Fahey, and R. G. Cosentino, Phosphine in the Venusian atmosphere: A strict upper limit from SOFIA GREAT observations, Geophysical Research Letters **49**, e2022GL098269 (2022).

[37] W. Bains, J. J. Petkowski, S. Seager, S. Ranjan, C. Sousa-Silva, P. B. Rimmer, Z. Zhan, J. S. Greaves, and A. M. S. Richards, Phosphine on Venus cannot be explained by conventional processes, Astrobiology **21**, 1277 (2021), demonstrates proposed metabolic pathways cannot explain Venus atmospheric chemistry.

[38] G. A. Schmidt and A. Frank, The Silurian hypothesis: would it be possible to detect an industrial civilization in the geological record?, International Journal of Astrobiology **18**, 142 (2019).

[39] J. A. Ball, The zoo hypothesis, Icarus **19**, 347 (1973).

[40] J. J. Petkowski, S. Seager, D. H. Grinspoon, W. Bains, S. Ranjan, P. B. Rimmer, *et al.*, Stability of amino acid backbones in concentrated sulfuric acid: implications for Venus cloud habitability, Astrobiology 10.1089/ast.2023.0082 (2024).

[41] S. Seager, J. J. Petkowski, *et al.*, Dipeptide stability in concentrated sulfuric acid: implications for Venus cloud biochemistry (2024), follow-up study on peptide bond stability under Venus cloud conditions.

[42] NASA NSSDCA, Venera 13 descent craft, `https://nssdc.gsfc.nasa.gov/nmc/spacecraft/display.action?id=1981-106D`, accessed 2026.

[43] R. G. Strom, G. G. Schaber, and D. D. Dawson, The global resurfacing of Venus, Journal of Geophysical Research: Planets **99**, 10899 (1994).

[44] J. S. Greaves, A. M. S. Richards, W. Bains, *et al.*, Phosphine gas in the cloud decks of Venus, Nature Astronomy **5**, 655 (2021), online first 2020, print 2021.

[45] NASA, VERITAS (venus emissivity, radio science, insar, topography, and spectroscopy), `https://science.nasa.gov/mission/veritas/` (), accessed 2026.

[46] ESA, EnVision mission, `https://www.esa.int/Science_Exploration/Space_Science/Envision` (), accessed 2026.

[47] J. P. Kennett and L. D. Stott, Abrupt deep-sea warming, palaeoceanographic changes and benthic extinctions at the end of the Palaeocene, Nature **353**, 225 (1991).

[48] K. L. C. Bell *et al.*, How little we've seen: A visual coverage estimate of the deep seafloor, Science Advances **11**, 10.1126/sciadv.adp8602 (2025).

[49] A. L. Melott, B. S. Lieberman, C. M. Laird, L. D. Martin, M. V. Medvedev, B. C. Thomas, J. K. Cannizzo, N. Gehrels, and C. H. Jackman, Did a gamma-ray burst initiate the late Ordovician mass extinction?, International Journal of Astrobiology **3**, 55 (2004).

[50] E. W. Cliver and W. F. Dietrich, The 1859 space weather event revisited: limits of extreme activity, Journal of Space Weather and Space Climate **3**, A31 (2013).

[51] R. Gomes, H. F. Levison, K. Tsiganis, and A. Morbidelli, Origin of the cataclysmic Late Heavy Bombardment period of the terrestrial planets, Nature **435**, 466 (2005).

[52] A. Wallner, J. Feige, N. Kinoshita, M. Paul, L. K. Fifield, R. Golser, M. Honda, U. Linnemann, H. Matsushi, S. Murgatroyd, L. Rutherford, and S. G. Tims, Recent

near-earth supernovae probed by global deposition of interstellar radioactive ^{60}Fe, Nature **532**, 69 (2016).

[53] S. H. Ambrose, Late Pleistocene human population bottlenecks, volcanic winter, and differentiation of modern humans, Journal of Human Evolution **34**, 623 (1998).

[54] G. P. Rightmire, Brain size and encephalization in early to Mid-Pleistocene *Homo*, American Journal of Physical Anthropology **124**, 109 (2004).

[55] NASA, DART (double asteroid redirection test) mission, https://science.nasa.gov/mission/dart/ (), accessed 2026.

[56] NASA, GRACE-FO (gravity recovery and climate experiment follow-on), https://gracefo.jpl.nasa.gov/ (), accessed 2026.

[57] L. W. Alvarez *et al.*, Search for hidden chambers in the pyramids, Science **167**, 832 (1970).

[58] The Nippon Foundation–GEBCO, Seabed 2030 project, https://seabed2030.org/, accessed 2026.

[59] NASA, MRO SHARAD (shallow radar), https://mars.nasa.gov/mro/mission/instruments/sharad/ (), accessed 2026.

[60] ESA, Mars Express MARSIS (mars advanced radar for subsurface and ionosphere sounding), https://www.esa.int/Science_Exploration/Space_Science/Mars_Express (), accessed 2026.

[61] K. J. Meech, R. Weryk, M. Micheli, *et al.*, A brief visit from a red and extremely elongated interstellar asteroid, Nature **552**, 378 (2017).

[62] P. Guzik, M. Drahus, K. Rusek, W. Waniak, G. Cannizzaro, and I. Pastor-Marazuela, Initial characterization of interstellar comet 2i/Borisov, Nature Astronomy **4**, 53 (2020), online first 2019.

[63] NASA Science, Comet 3I/ATLAS facts and FAQs (2025), third confirmed interstellar object, discovered 2025-07-01.

[64] v. Ivezić, S. M. Kahn, J. A. Tyson, *et al.*, LSST: From science drivers to reference design and anticipated data products, The Astrophysical Journal **873**, 111 (2019).

[65] Vera C. Rubin Observatory, Rubin observatory issues first scientific alerts (2026), first batch of LSST scientific alerts issued 2026-02-25.

[66] C. Snodgrass and G. H. Jones, The European Space Agency's Comet Interceptor lies in wait, Nature Communications **10**, 5418 (2019).